

Navya Jain

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WORK EXPERIENCE

Ritchie Bros (acquired Veritread in March 2026), Boston (Remote) July 2024 - Present
Machine Learning Engineer II (Oct 2025-Present)

- Spearheaded an AI bidding agent using **AWS Bedrock AgentCore**, **Strands SDK**, and **Claude Sonnet** to orchestrate pricing and bidding workflows, reducing manual bid processing time by 85%
- Built **end-to-end** machine learning models with **Pytorch** and **SKLearn** (regressors) to generate price estimations for transporting heavy hauls, increasing the overall quarterly revenue by 20%
- Developed a LinUCB-based contextual bandit system in **reinforcement learning**, improving automated price accuracy by 35% while ensuring compliance with business-fairness policies
- Engineered **100+ time-series features** and a **LightGBM confidence model** to score prediction uncertainty, enabling business stakeholders to set risk thresholds on automated quotes
- Deployed ML pricing models on **Amazon SageMaker AI** and made a REST API for generating predictions, pointing to an existing **EKS** (Kubernetes) service to handle 1,037,000 monthly requests
- Conducted **MLOps** with **MLFlow** for experiment tracking and deployed cron jobs to synchronize production data between **AWS S3** and **SSMS**
- Maintained observability with **AWS CloudWatch** by tuning metrics, logs, dashboards, and alarms, performing monthly monitoring upgrades to ensure reliable system performance

Software Engineer (Machine Learning) (July 2024-Sept 2025)

- Developed internal tools for price forecasting using **gradient boosting** methods in **SKLearn** for client-specific freight movement, directly impacting business decisions
- Tested **.NET 6** and **C# REST APIs** for internal pricing services, enabling reliable **SQL**-based data access
- Employed **CI/CD** methodologies with **CircleCI** to deploy code to dev, QA, and prod environments seamlessly

Boston University, Boston

January 2023 - January 2024

Head Teaching Assistant for Foundations of Data Science III

- Led discussion sessions and created assignments on concepts in probability, statistics, and data science
- Explained topics like Bayesian statistics, optimization methods, and sampling, along with subsequent implementations using Python libraries like pandas, numpy, scipy, and scikit-learn

EDUCATION

Boston University

Massachusetts, USA

Master of Science, Artificial Intelligence

September 2022 - January 2024

Guru Gobind Singh Indraprastha University

Delhi, India

Bachelor of Technology, Electrical and Electronics Engineering

August 2018 - June 2022

PROJECTS

Multimodal Vision-Language Model (PaliGemma) from Scratch

- Implemented a **PyTorch multimodal architecture** by implementing a **SigLIP Vision Transformer**, Gemma decoder, and a projection bridge, with custom RoPE, RMSNorm, attention, and KV caching
- Loaded and aligned **PaliGemma** pre-trained weights to validate the implementation, achieving accurate image-conditioned text generation on real-world examples

Semantic Book Recommendation with Two-Tower CF Architecture

- Created an **agentic** book recommender in Pytorch with over 2M books and 250,000 user ratings to translate natural-language preferences into personalized results using **two-tower retrieval** and **LLM reranking**
- Combined **neural collaborative filtering** with BM25 and dense semantic search (**sentence-transformers**) over review passages to surface books matching user taste and emotional intent beyond keyword match

MLOps and DevOps with Customer Churn Prediction

- Quantified a \$17M revenue gap via churn analysis, framing customer retention as a key business lever, and trained **ANNs**, **ensemble**, and linear models using k-fold validation and **MLflow**
- Engineered a reproducible MLOps pipeline with **DVC**, **GitHub Actions**, and **AWS CloudWatch**, deploying a **Streamlit** app for real-time churn prediction and batch retraining on AWS

SKILLS

- **Programming Languages:** Python, SQL, TypeScript, JavaScript, C++, Bash
- **ML/AI:** PyTorch, scikit-learn, TensorFlow, NumPy, pandas, Hugging Face, LangChain, LLMs, RAG, MLflow
- **Cloud / MLOps / Tools:** AWS, Bedrock, Docker, EKS, DVC, Git, CircleCI, Spark, Strands, JIRA

OTHER EXPERIENCE

Realix.ai, Boston (Remote)

November 2024 - January 2025

Data Scientist (Contract)

- Designed a GPT-4 Turbo RAG pipeline with LlamaIndex to convert natural language into executable SQL, conducted QA with varied question phrasings for optimization, achieving 85% accuracy across query formats

Nawaazish Crew (Charity), Delhi, India

February 2020 - January 2021

Cofounder and President

- Co-founded a nonprofit in India; led 45 volunteers and led COVID relief for 30,000 families
- Developed a clustering-based segmentation model to improve volunteer onboarding